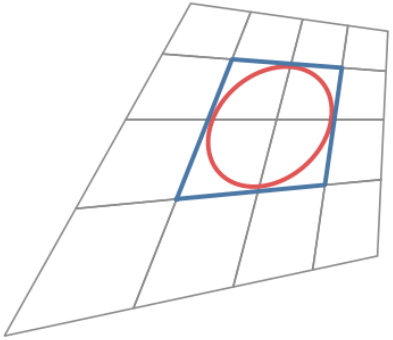
Industrial Vision

Lecture 8 - The projective plane

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What a photograph destroys, and the one thing it keeps





- The bands of the facade are **parallel on the building** and **concurrent in the image**. The seven drawn here were found by Canny and the Hough transform of **L03**, and their common point sits just outside the frame with a residual below two pixels.
- A **square tile is not a square** and a **circular pattern is not a circle**. Length, angle, ratio of lengths and parallelism are all gone. What survives? **straight edge images as a straight edge**.
- Therefore, a **projective transformation carries straight lines to straight lines**, and projective geometry studies what is invariant under such mappings.

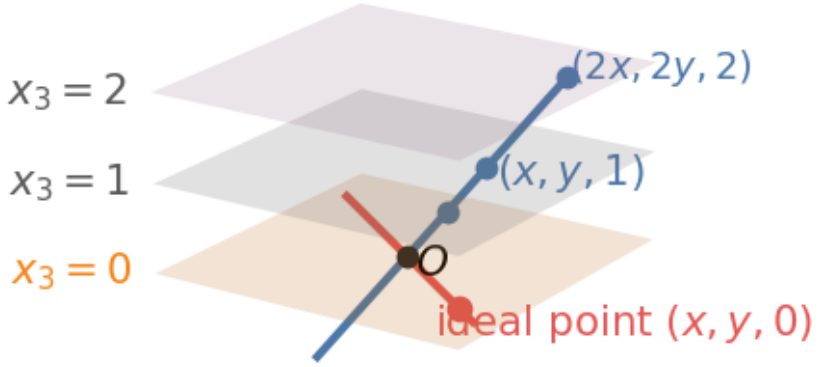
Homogeneous coordinates

A point $(x, y) \in \mathbb{R}^2$ is represented by the 3-vector $(kx,\ ky,\ k)^\top$ for any $k \neq 0$, and (x, y) is recovered as $(kx/k,\ ky/k)$. All such vectors are **equivalent**, so a point of $\mathbb{P}^2$ is an **equivalence class**:

$$\mathbf{x} = (x_1, x_2, x_3)^\top \in \mathbb{P}^2 \ \equiv\ \left(\tfrac{x_1}{x_3},\ \tfrac{x_2}{x_3}\right)^\top \in \mathbb{R}^2.$$

The zero vector is **excluded**, so $\mathbb{P}^2 = \left(\mathbb{R}^3 \setminus \{\mathbf{0}\}\right)/\sim$. It is a class by itself, since no scaling reaches it or leaves it, and in the ray model it is the origin, which is not a ray.

Stack the planes $x_3 = k$ in $\mathbb{R}^3$. A point of $\mathbb{P}^2$ is then a **line through the origin**, and it meets the plane $x_3 = 1$ in the corresponding Euclidean point.



A ray **parallel** to the plane $x_3 = 1$ never meets it, and its third coordinate is zero. So $(x, y, 0)^\top$ is a legitimate element of $\mathbb{P}^2$ with no Euclidean counterpart, since $(x/0,\ y/0)$ is infinite. It is called a **point at infinity**, or an **ideal point**.

$\mathbb{P}^2$ is $\mathbb{R}^2$ **together with its points at infinity**, and infinity is now written in finite coordinates.

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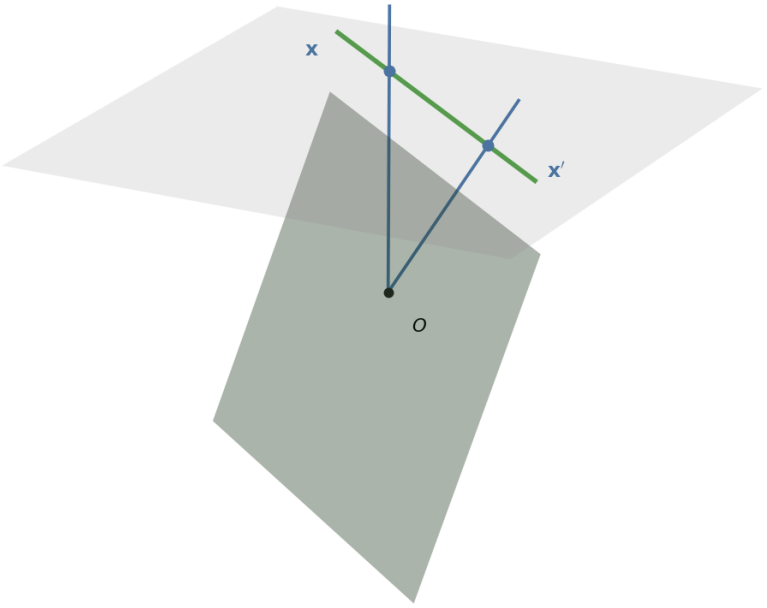
Lines are 3-vectors and incidence is a dot product

A line in the plane is $ax + by + c = 0$, and different triples (a, b, c) give different lines, so a line is naturally the 3-vector $\mathbf{l} = (a, b, c)^\top$. It is homogeneous for the same reason a point is, since (a, b, c) and $k(a, b, c)$ describe the same line, and $(0, 0, 0)^\top$ is excluded here too: $0x + 0y + 0 = 0$ holds for **every** point and so cuts out no line.

The point (x, y) lies on $\mathbf{l}$ if and only if $ax + by + c = 0$, that is $(x, y, 1)\,(a, b, c)^\top = 0$. For any $k \neq 0$ we have $(kx, ky, k)\,\mathbf{l} = k\,(x, y, 1)\,\mathbf{l} = 0$, so the relation is well defined on equivalence classes:

$$\boxed{\mathbf{x}^\top \mathbf{l} = 0}$$

This is a scalar product, so $\mathbf{x}^\top \mathbf{l} = \mathbf{l}^\top \mathbf{x}$. Points and lines are the same kind of object, distinguished only by which side of the product they sit on.



Degrees of freedom. A point has two, being x and y. A line also has two, being the independent ratios $\{a : b : c\}$. Three homogeneous coordinates on each side, one of them lost to scale.

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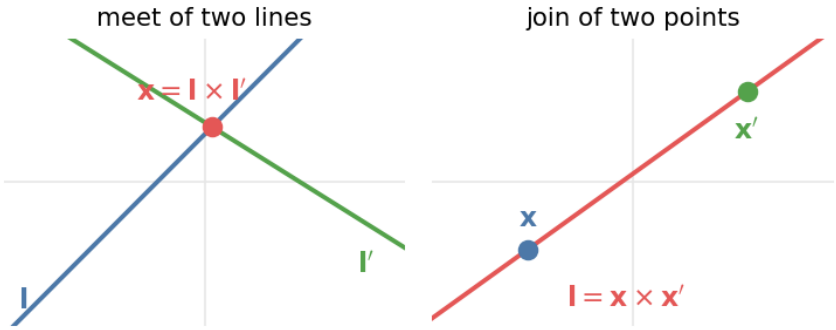
Cross product of lines and points

The intersection of two lines. The meet of two **distinct** lines $\mathbf{l}$ and $\mathbf{l}'$ is $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$.

- The triple scalar product gives $\mathbf{l} \cdot (\mathbf{l} \times \mathbf{l}') = \mathbf{l}' \cdot (\mathbf{l} \times \mathbf{l}') = 0$, which reads $\mathbf{l}^\top \mathbf{x} = \mathbf{l}'^\top \mathbf{x} = 0$. The point $\mathbf{x}$ therefore lies on both lines.

The line through two points. The join of two **distinct** points $\mathbf{x}$ and $\mathbf{x}'$ is $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.

- The same identity with the roles exchanged, since $\mathbf{x} \cdot (\mathbf{x} \times \mathbf{x}') = \mathbf{x}' \cdot (\mathbf{x} \times \mathbf{x}') = 0$ gives $\mathbf{x}^\top \mathbf{l} = \mathbf{x}'^\top \mathbf{l} = 0$.



The second proof is the first with the words swapped, which is a direct consequence of the homogeneous representation.

Worked example: The lines $x = 1$ and $y = 1$ are $\mathbf{l} = (1, 0, -1)^\top$ and $\mathbf{l}' = (0, 1, -1)^\top$, and $\mathbf{l} \times \mathbf{l}' = (1, 1, 1)^\top$, which is the point $(1, 1, 1)$. The cross product returns $\mathbf{0}$ **exactly** when the two arguments are the same line, or the same point, and $\mathbf{0}$ is not an element of $\mathbb{P}^2$ — which is why both statements are made for distinct arguments.

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Ideal points and the line at infinity

Take two **parallel** lines, $\mathbf{l} = (a, b, c)^\top$ and $\mathbf{l}' = (a, b, c')^\top$ with $c \neq c'$. Their meet is

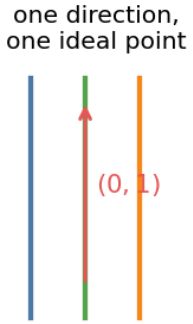
$$\mathbf{l} \times \mathbf{l}' = (c' - c)\,(b, -a, 0)^\top \;\sim\; (b, -a, 0)^\top.$$

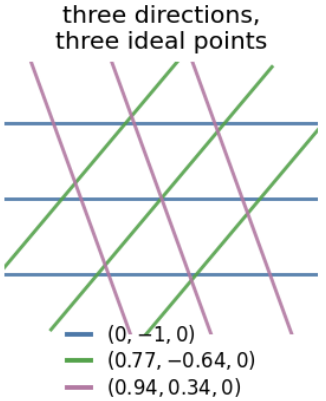
The third coordinate is **zero**, so the intersection is an ideal point. Parallel lines do meet, and they meet at infinity. In Euclidean geometry this is a special case to be excluded.

$$x = 1, \;\; x = 2 \implies \mathbf{l} = (1, 0, -1)^\top, \; \mathbf{l}' = (1, 0, -2)^\top,$$

$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = (0, 1, 0)^\top,$$

the point at infinity in the direction of the y-axis, which is the direction along which both lines run.





Interpretation. In inhomogeneous terms $(b, -a)^\top$ is orthogonal to the line normal $(a, b)^\top$, so it is **tangent** to the line and represents its **direction**. Every line of that direction, whatever its c, meets at the same ideal point.

The line at infinity. The ideal points $(x_1, x_2, 0)^\top$ all satisfy $\mathbf{x}^\top \mathbf{l} = 0$ with $\mathbf{l}_\infty = (0, 0, 1)^\top$, so they are **collinear**.

$\boxed{\mathbf{l}_\infty = (0, 0, 1)^\top}$ the line at infinity is the **set of directions of lines in the plane**

The duality principle

Everything so far has arrived in pairs. A point is a 3-vector and so is a line. Incidence is $\mathbf{x}^\top \mathbf{l} = 0$ and reads the same in both directions. The meet of two lines and the join of two points are the same cross product with the same proof.

> **Duality principle.** To any theorem of two-dimensional projective geometry there corresponds a **dual theorem**, obtained by interchanging the roles of points and lines. It is not necessary to prove the dual once the original is proved, because **the proof of the dual is the dual of the proof**.

This is not a new idea for this class. The Hough transform of **L03** was the same move: a point of the image is a **sinusoid** in the parameter space, and points that are collinear in one space become curves that are concurrent in the other. Only the name and the generality are new.

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Conics, five degrees of freedom, and five points

A conic is a 2nd order polynomial:

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

Homogenising with $x \to x_1/x_3$ and $y \to x_2/x_3$ and multiplying by x_3^2 gives,

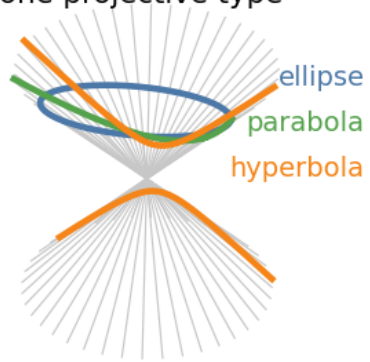
$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

that is

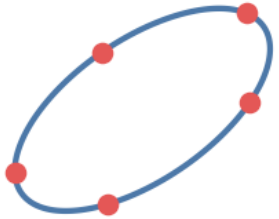
$$\mathbf{x}^\top C\mathbf{x} = 0, \quad C = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$

The matrix is **symmetric**, and homogeneous, since scaling it does not change the curve. A conic therefore has **five degrees of freedom**, being six entries of a symmetric matrix less one for scale.

one projective type



five points, one null vector



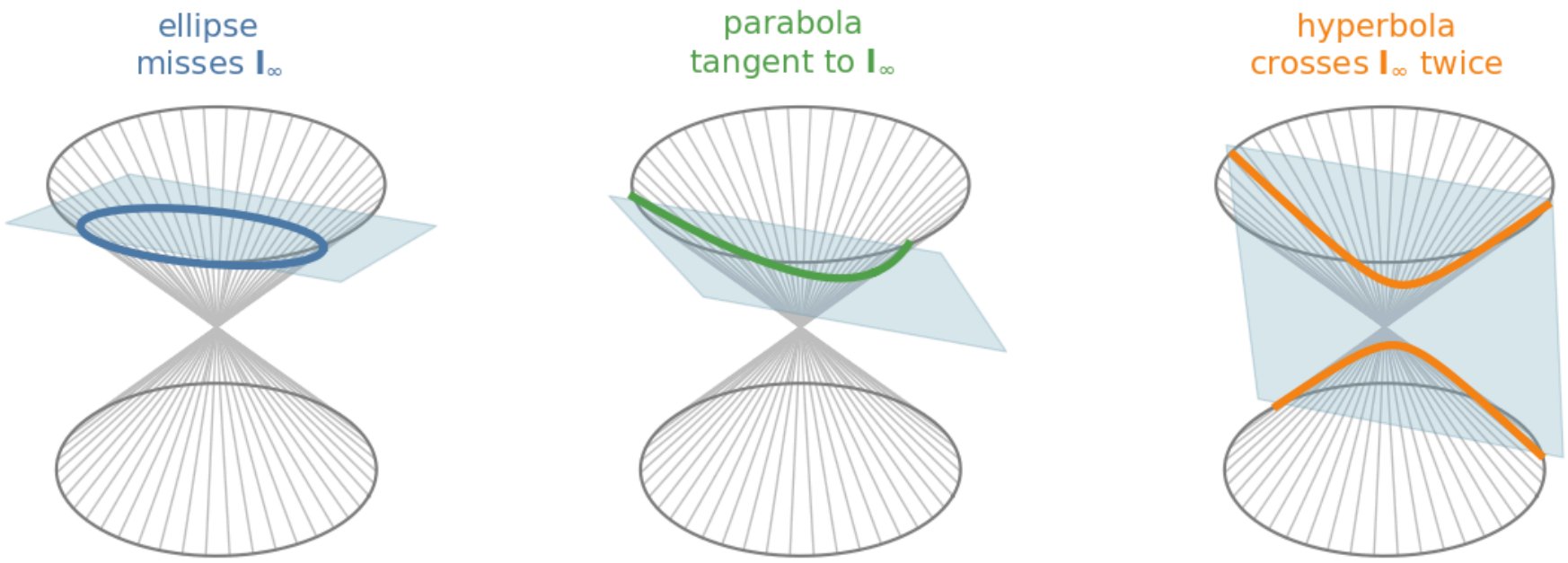
Each point on the conic gives one linear constraint,

$$\begin{bmatrix} x_i^2 & x_iy_i & y_i^2 & x_i & y_i & 1 \end{bmatrix} \mathbf{c} = 0,$$

with $\mathbf{c} = (a, b, c, d, e, f)^\top$. Stacking **five** such rows gives a 5×6 matrix, and the conic is its **null vector**.

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The cutting plane and the three types



A plane cutting $x^2 + y^2 = z^2$ produces an ellipse, parabola, or hyperbola according to how the plane meets the line at infinity.

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Tangency and the dual conic

The tangent to C at a point $\mathbf{x}$ of C is $\mathbf{l} = C\mathbf{x}$.

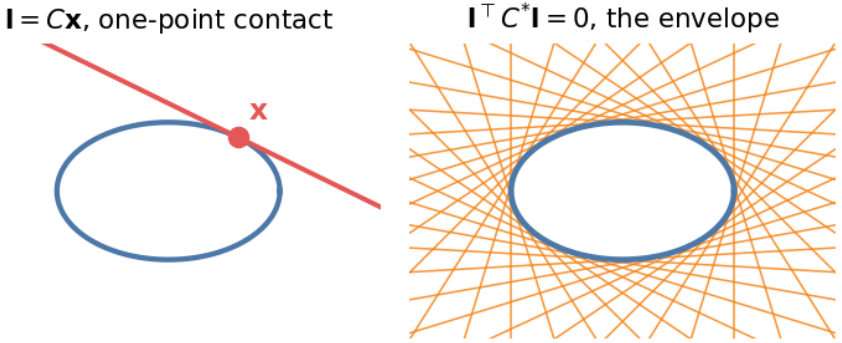
- The line $\mathbf{l} = C\mathbf{x}$ passes through $\mathbf{x}$, since $\mathbf{l}^\top \mathbf{x} = \mathbf{x}^\top C\mathbf{x} = 0$. If it has one-point contact it is the tangent.

Otherwise it meets the conic again at some $\mathbf{y}$, so $\mathbf{y}^\top C\mathbf{y} = 0$ and $\mathbf{x}^\top C\mathbf{y} = \mathbf{l}^\top \mathbf{y} = 0$; then $(\mathbf{x} + \alpha\mathbf{y})^\top C(\mathbf{x} + \alpha\mathbf{y}) = 0$ for every α, the whole line lies on C, and C is degenerate.

By duality there must also be a conic that constrains **lines**. A line tangent to C satisfies

$$\mathbf{l}^\top C^* \mathbf{l} = 0, \qquad C^* = C^{-1}$$

up to scale, for non-singular C.



Derivation. At $\mathbf{x}$ on C the tangent is $\mathbf{l} = C\mathbf{x}$, so $\mathbf{x} = C^{-1}\mathbf{l}$. Substituting into $\mathbf{x}^\top C\mathbf{x} = 0$ gives $\mathbf{l}^\top C^{-1}\mathbf{l} = 0$, using $C^{-\top} = C^{-1}$ because C is symmetric.

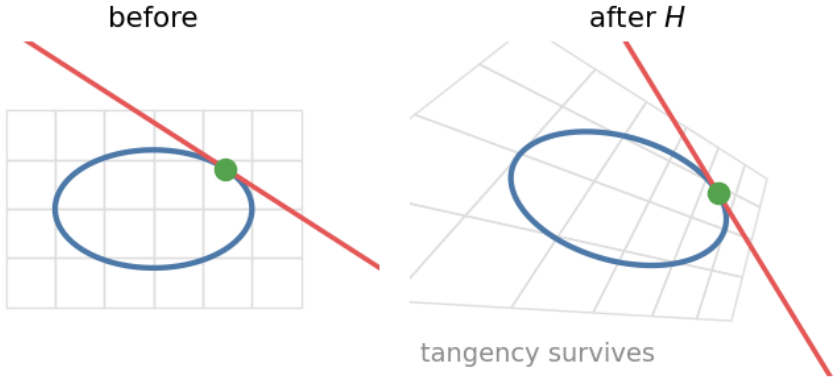
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How H acts on a point, a line and a conic

A **projectivity** or **homography** is an invertible map $h : \mathbb{P}^2 \to \mathbb{P}^2$ under which three points are collinear if and only if their images are. Equivalently, h is a projectivity if and only if there is a non-singular 3×3 matrix H with $h(\mathbf{x}) = H\mathbf{x}$.

- Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ lie on $\mathbf{l}$, so $\mathbf{l}^\top \mathbf{x}_i = 0$. For non-singular H we have $\mathbf{l}^\top H^{-1} H\mathbf{x}_i = 0$, so the points $H\mathbf{x}_i$ all lie on the line $H^{-\top}\mathbf{l}$ and collinearity is preserved.

Incidence is preserved by construction, since $\mathbf{l}'^\top \mathbf{x}' = \mathbf{l}^\top H^{-1} H\mathbf{x} = \mathbf{l}^\top \mathbf{x}$.



For a conic, substitute $\mathbf{x} = H^{-1}\mathbf{x}'$ into $\mathbf{x}^\top C\mathbf{x} = 0$ to obtain the quadratic form $\mathbf{x}'^\top \left[H^{-1}\right]^\top C H^{-1}\mathbf{x}'$.

For a dual conic, substitute $\mathbf{l}' = H^{-\top}\mathbf{l}$ into $\mathbf{l}^\top C^*\mathbf{l} = 0$ to obtain the quadratic form $(H^\top \mathbf{l}')^\top C^* (H^\top \mathbf{l}') = \mathbf{l}'^\top (H\mathbf{C}^* H^\top)\mathbf{l}'$.

$$\boxed{\mathbf{x}' = H\mathbf{x}}, \qquad \boxed{\mathbf{l}' = H^{-\top}\mathbf{l}}, \qquad \boxed{C' = H^{-\top} C H^{-1}}, \qquad \boxed{C^{*\prime} = H C^* H^\top}$$

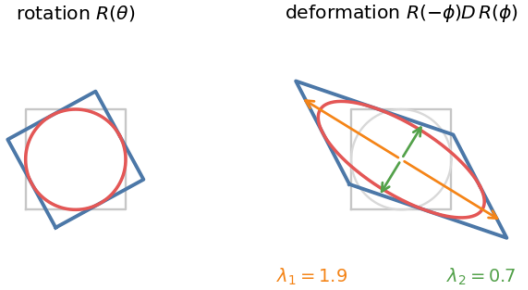
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The hierarchy of transformations

group	matrix	action on $\mathbf{x} = (\tilde{\mathbf{x}}, 1)^\top$	dof	distortion	invariants
Euclidean	$\begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$	$\begin{bmatrix} R\tilde{\mathbf{x}} + \mathbf{t} \\ 1 \end{bmatrix}$	3		length, angle, area
similarity	$\begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$	$\begin{bmatrix} sR\tilde{\mathbf{x}} + \mathbf{t} \\ 1 \end{bmatrix}$	4		ratio of lengths, angle, the circular points I, J (L11)
affine	$\begin{bmatrix} A & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$	$\begin{bmatrix} A\tilde{\mathbf{x}} + \mathbf{t} \\ 1 \end{bmatrix}$	6		parallelism, ratio of areas, $\mathbf{l}_\infty$ **fixed** (L11)
projective	$\begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix}$	$\begin{bmatrix} A\tilde{\mathbf{x}} + \mathbf{t} \\ \mathbf{v}^\top\tilde{\mathbf{x}} + v \end{bmatrix}$	8		collinearity, concurrency, order of contact, the **cross ratio**

$\tilde{\mathbf{x}} = (x_1, x_2)^\top$ is the inhomogeneous part. The bottom row $(\mathbf{0}^\top, 1)$ of the first three groups leaves $x_3 = 1$ untouched, so the Euclidean point is read off directly and **no division is ever**

- Each group is a special case of the next and inherits the invariants above it. A similarity is fixed by **two** point correspondences, an affinity by **three**, a projectivity by **four** with no three collinear.
- By the SVD, $A = R(\theta)\, R(-\phi)\, D\, R(\phi)$: an affinity is a rotation, a scaling by λ_1, λ_2 along orthogonal directions, and a rotation back. This is the eigen-picture of **L05**, and it is why a circle becomes an ellipse with those axes.

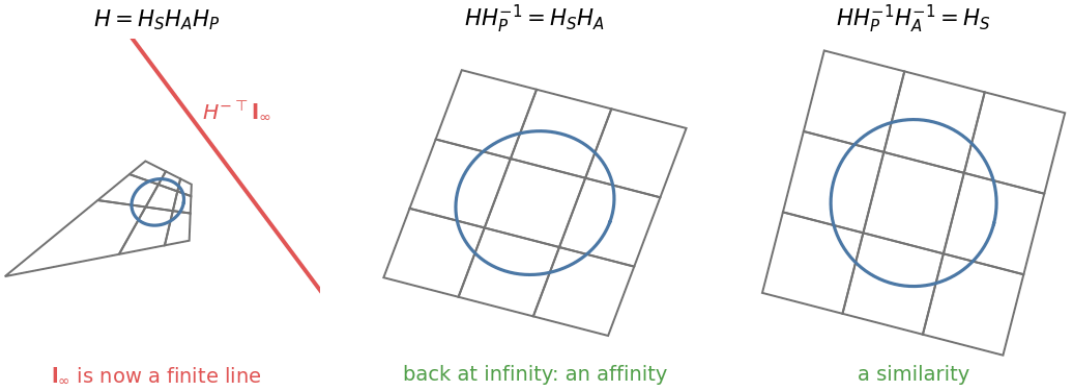


The decomposition $H = H_S H_A H_P$

A projectivity factors into a chain in which each matrix is higher in the hierarchy than the last:

$$H = H_S H_A H_P = \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^\top & v \end{bmatrix} = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix}$$

with $A = sRK + \mathbf{t}\mathbf{v}^\top$ and K upper triangular normalised so that $\det K = 1$. The factorisation is valid provided $v \neq 0$, and it is unique if s is taken positive.



H_P is the only factor that moves $\mathbf{l}_\infty$, and it carries it to the finite line drawn in red. Undo H_P and the transformation is an affinity; undo H_A as well and it is a similarity.

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One dimension down: $\mathbb{P}^1$ and the cross ratio

A point on a line is $\bar{\mathbf{x}} = (x_1, x_2)^\top$, and $x_2 = 0$ is its ideal point. A projective transformation of the line is a 2×2 homogeneous matrix $\bar{\mathbf{x}}' = H_{2\times 2}\bar{\mathbf{x}}$ with **three degrees of freedom** and is determined by three points.

$$\text{Cross}(\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \bar{\mathbf{x}}_3, \bar{\mathbf{x}}_4) = \frac{|\bar{\mathbf{x}}_1\bar{\mathbf{x}}_2|\ |\bar{\mathbf{x}}_3\bar{\mathbf{x}}_4|}{|\bar{\mathbf{x}}_1\bar{\mathbf{x}}_3|\ |\bar{\mathbf{x}}_2\bar{\mathbf{x}}_4|}$$

$$|\bar{\mathbf{x}}_i\bar{\mathbf{x}}_j| = \det \begin{bmatrix} x_{i1} & x_{j1} \\ x_{i2} & x_{j2} \end{bmatrix}$$

The motivation is elimination. A length is not invariant under a projectivity, and neither is a ratio of lengths, which dies at the projective level. A ratio **of ratios** survives, and the count agrees: a projectivity of the line is fixed by three points, so the first invariant quantity involves four. A plane projectivity induces a one-dimensional projectivity on **every line of the plane**, which is why this matters for images.

every length changes, the cross ratio does not

one pencil, three transversals



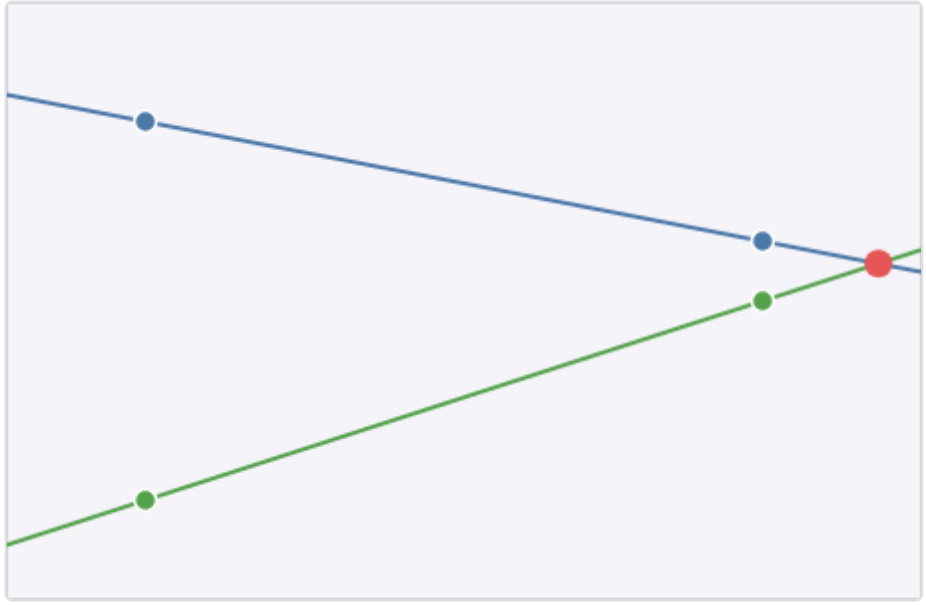
line	$\lvert x_1 x_2\rvert$	ratio	cross ratio
1	1.700	0.500	0.2875
2	0.734	0.413	0.2875
3	0.681	0.587	0.2875

- The value does not depend on which representative of each class is used, since the scales cancel between numerator and denominator.
- With each point scaled so that $x_2 = 1$, the determinant is the **signed distance** from $\bar{\mathbf{x}}_i$ to $\bar{\mathbf{x}}_j$, so this really is a ratio of ratios of lengths.
- The definition remains valid if one of the four points is **ideal**, and it is **invariant under every projectivity of the line**. That proof is left as an exercise.

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Demonstration

meet and join | the cross ratio | the hierarchy



l	(0.190, -0.982, 45.605)
l′	(0.307, 0.952, -259.417)
l × l′	(-1.000, -0.300, **-0.0023**)

Euclidean intersection (438.1, 131.3). Drag the two lines toward parallel and watch the third coordinate fall.

make parallel | reset

- Source-to-target vs target-to-source transformations.

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Questions

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Summary

- A photograph destroys length, angle, ratio of lengths and parallelism. It preserves **straightness**, and projective geometry is the geometry of that one invariant.
- A point of $\mathbb{P}^2$ is an equivalence class (kx, ky, k), equivalently a **ray through the origin** of $\mathbb{R}^3$. Rays parallel to the plane $x_3 = 1$ give the **ideal points** $(x, y, 0)$, so infinity is written in finite coordinates.
- Points and lines are both 3-vectors joined by $\mathbf{x}^\top \mathbf{l} = 0$. The **meet** of two lines and the **join** of two points are the same cross product with the same proof, which is the content of the **duality principle**.
- Parallel lines (a, b, c) and (a, b, c') meet at $(b, -a, 0)$, an ideal point that records the direction. All ideal points lie on $\mathbf{l}_\infty = (0, 0, 1)^\top$, so **the line at infinity is the set of directions in the plane**.
- A conic is $\mathbf{x}^\top C \mathbf{x} = 0$ with C symmetric and five degrees of freedom, fitted to **five points** as a null vector. Its tangent at $\mathbf{x}$ is $\mathbf{l} = C\mathbf{x}$ and its **dual** $C^* = C^{-1}$ constrains tangent lines.
- Under $\mathbf{x}' = H\mathbf{x}$ we have $\mathbf{l}' = H^{-\top}\mathbf{l}$ and $C' = H^{-\top} C H^{-1}$.
- The **hierarchy**, of 3, 4, 6 and 8 degrees of freedom, is a ladder of what each group destroys, and $H = H_S H_A H_P$ isolates H_P as the only factor that moves $\mathbf{l}_\infty$.
- On a line the **cross ratio** of four points is invariant under every projectivity. It is a ratio of ratios of lengths, and it is the first invariant that survives.

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References

1. Hartley, Richard, and Andrew Zisserman. Multiple view geometry in Industrial Vision. Vol. 2. No. 3. Cambridge: Cambridge university press, 2003. See Ch. 2, "Projective geometry and transformations of 2D".